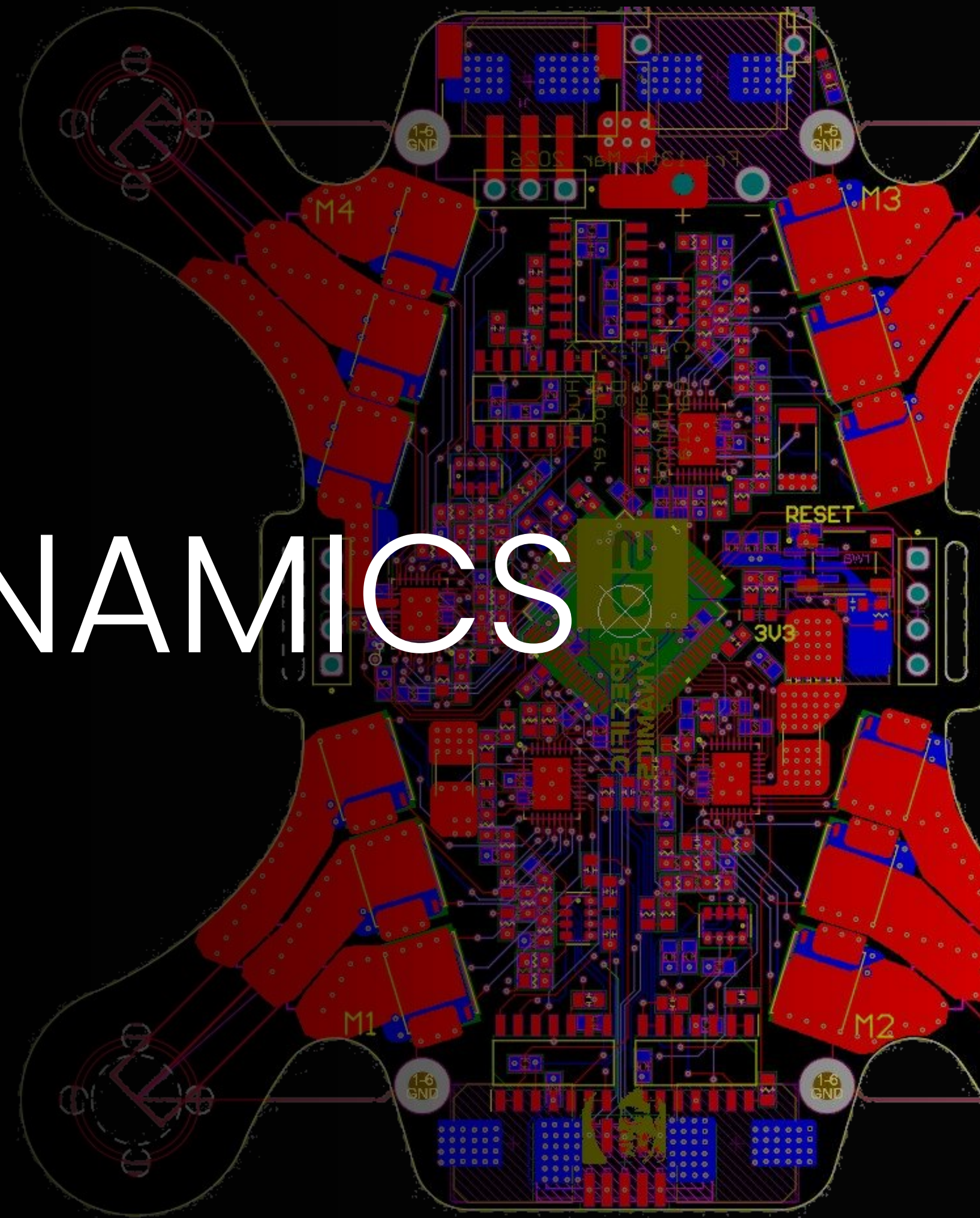


SPECIFIC DYNAMICS

A Higher Purpose.



THE PROBLEM

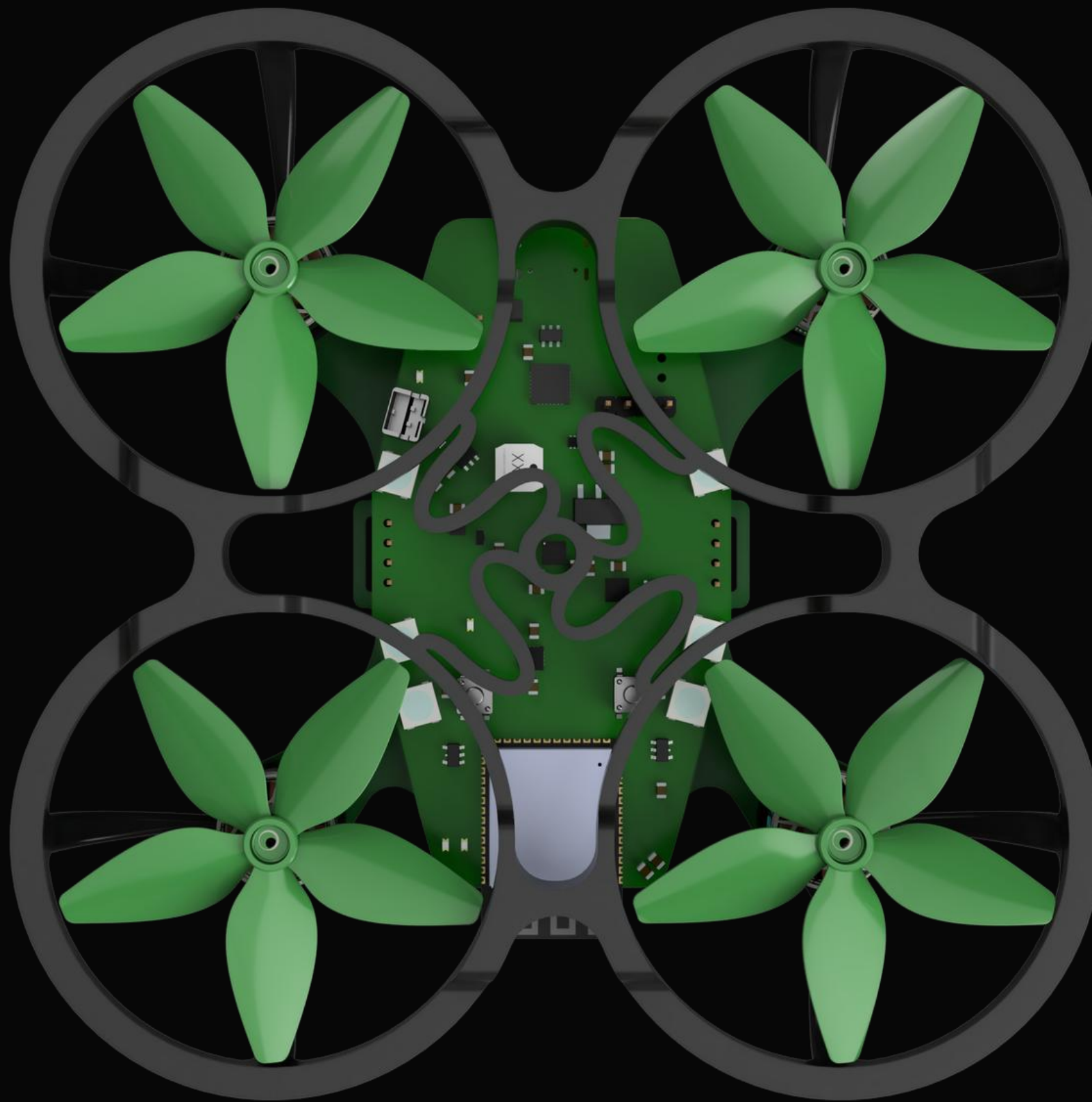
- Commercial micro-drones: \$300+ with closed ecosystems
- Limited customization for specialized tasks
- Long development timelines for purpose-built platforms
- No affordable, modular, hackable sub-250 g platform exists

F17-HUMANITARIAN HUMMINGBIRD

A 145 g microdrone platform where the printed circuit board itself serves as the structural airframe, eliminating the need for a traditional frame. The system integrates a custom brushless motor controller, PID-stabilized flight controller, and ESP-NOW wireless control link, all designed and built from scratch. The result is a compact, low-cost, and fully modular proof-of-concept platform that can be adapted for general-purpose UAV applications including aerial sensing, imaging, and small payload delivery.



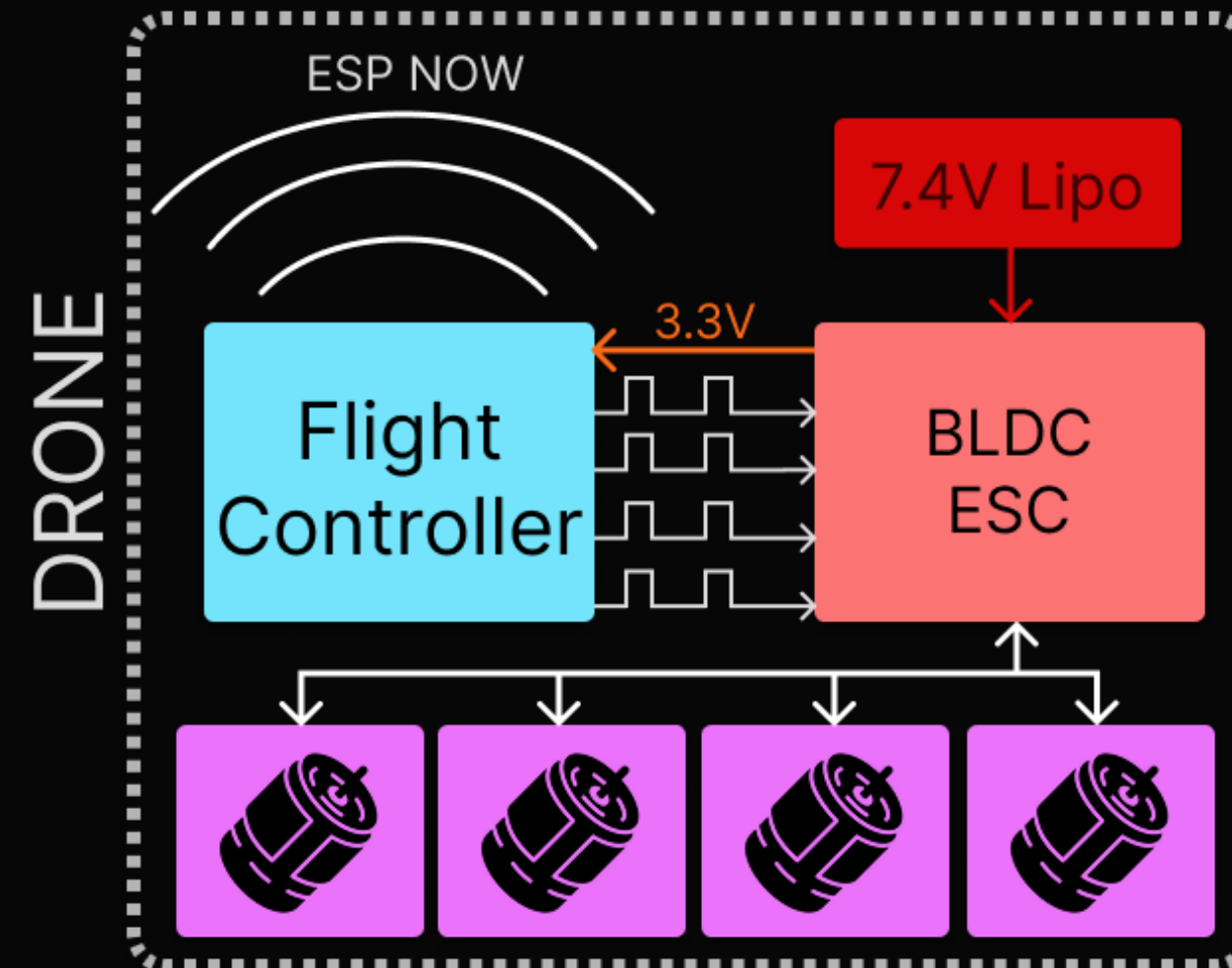
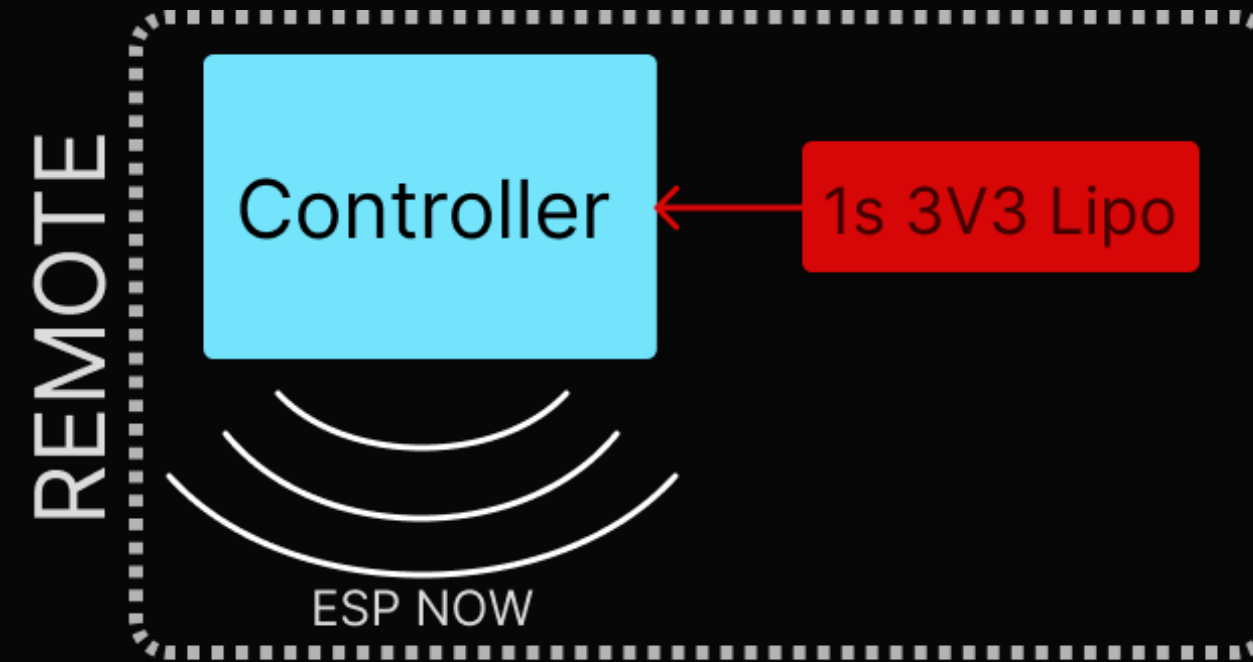
LET'S SEE THIS SH!T
FLY



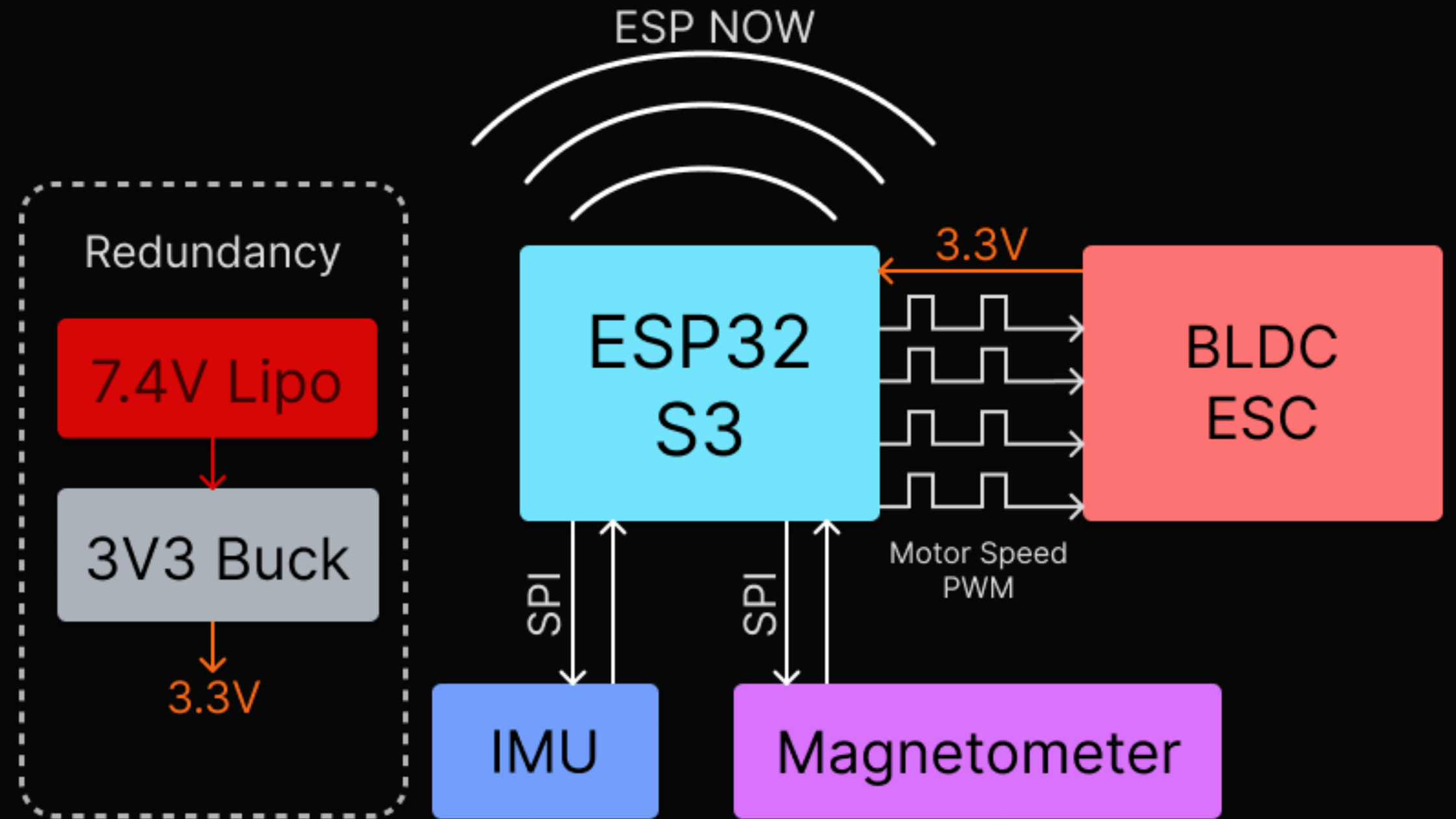
SYSTEM ARCHITECTURE



HIGH-LEVEL SYSTEM OVERVIEW

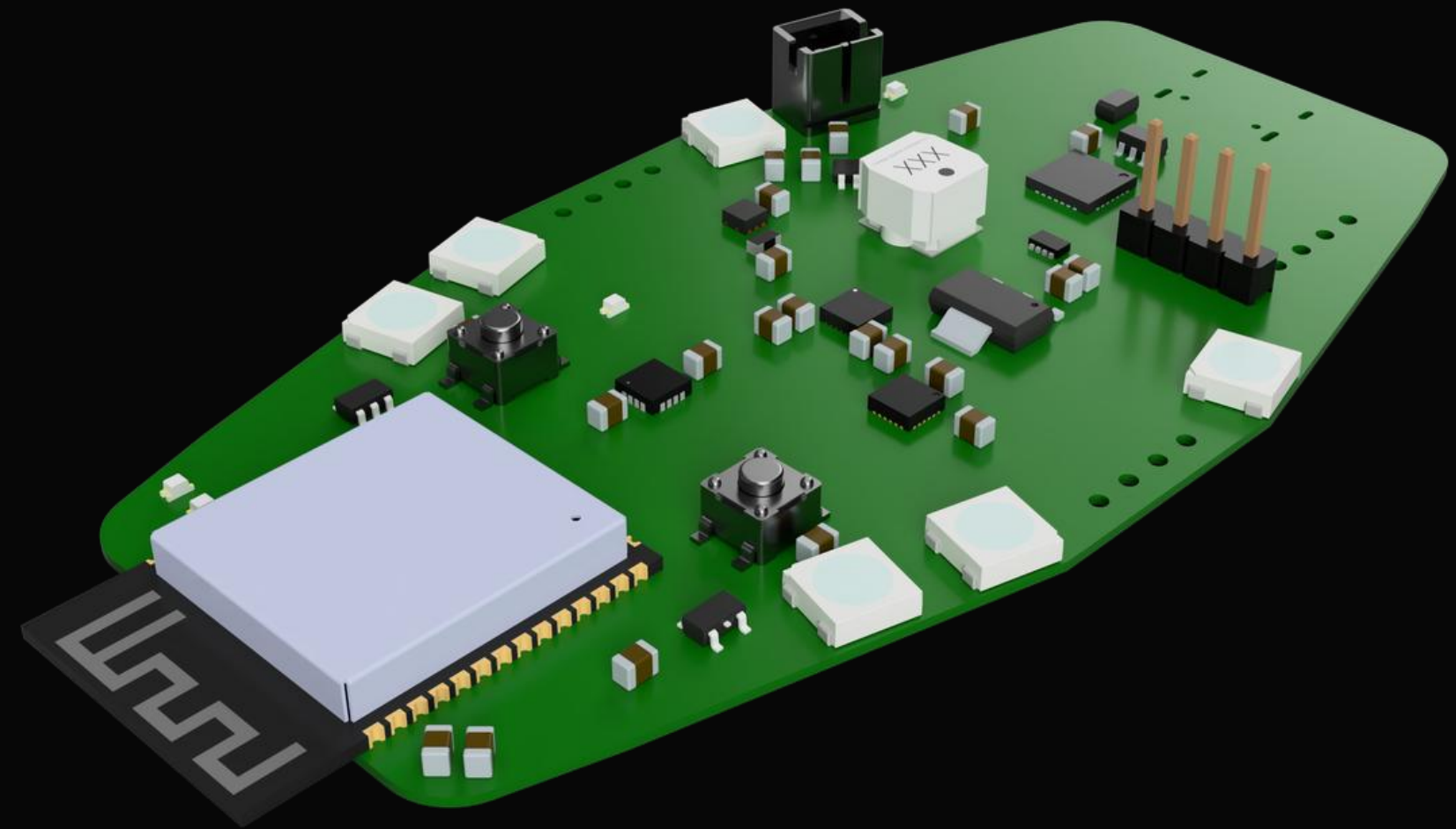


AVIONICS FLIGHT CONTROLLER

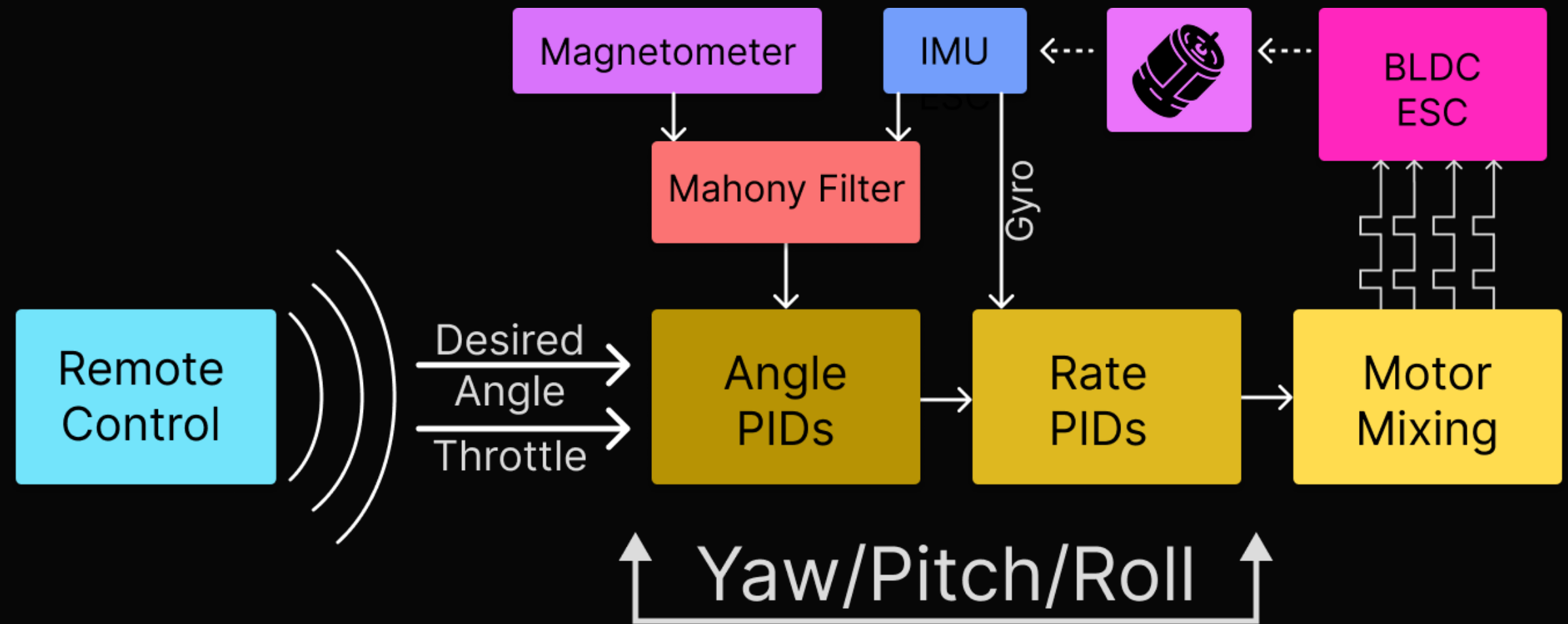


PID/AVIONICS FLIGHT CONTROLLER

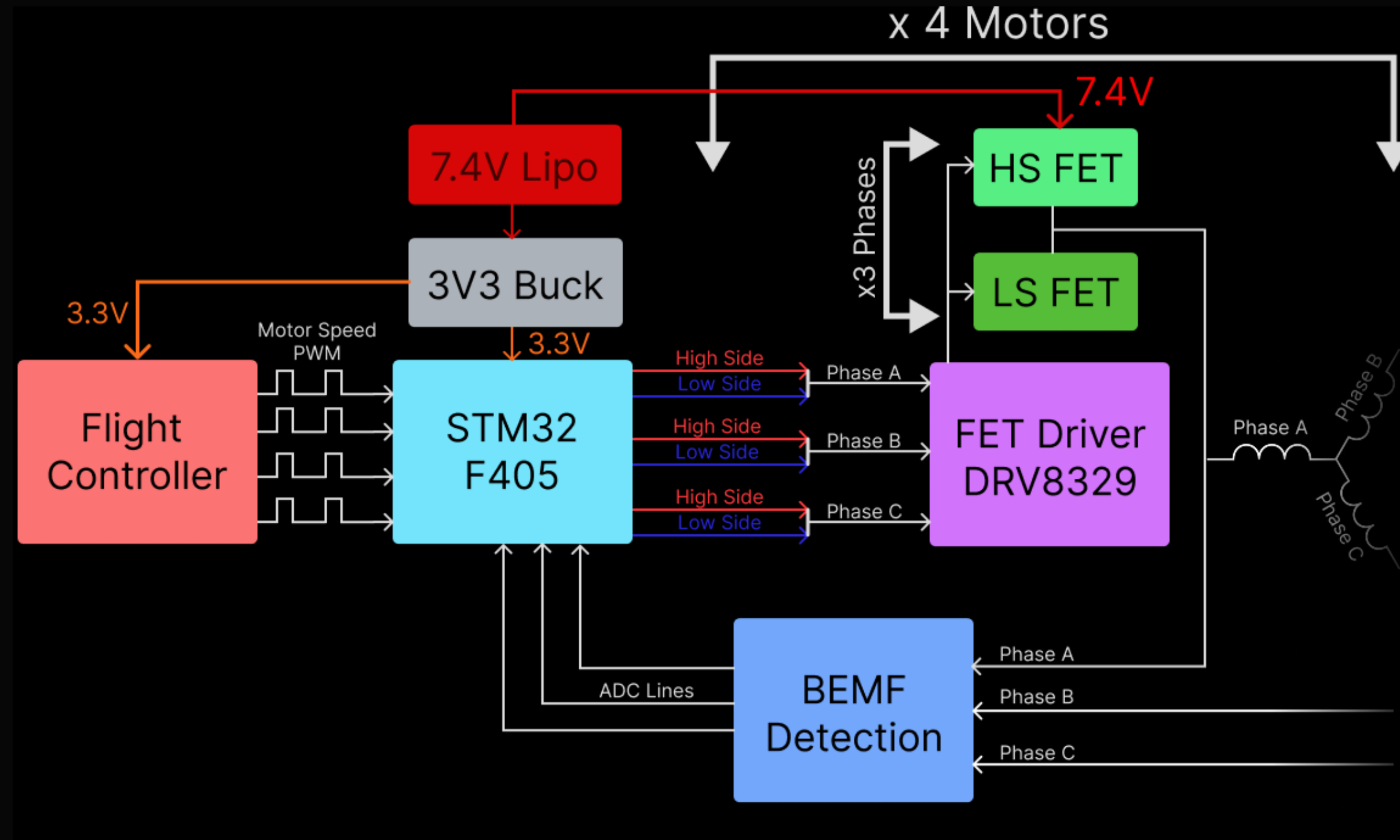
- 6 cascaded loops - outer angle + inner rate for each axis (roll, pitch, yaw)
- 1 kHz loop rate
- 9-axis sensor fusion - Mahony filter (gyro + accel + magnetometer)
- Ziegler-Nichols tuned, empirically refined in flight
- 4-state failsafe - 1 s timeout → motors off → must re-arm



AVIONICS FLIGHT CONTROLLER FIRMWARE

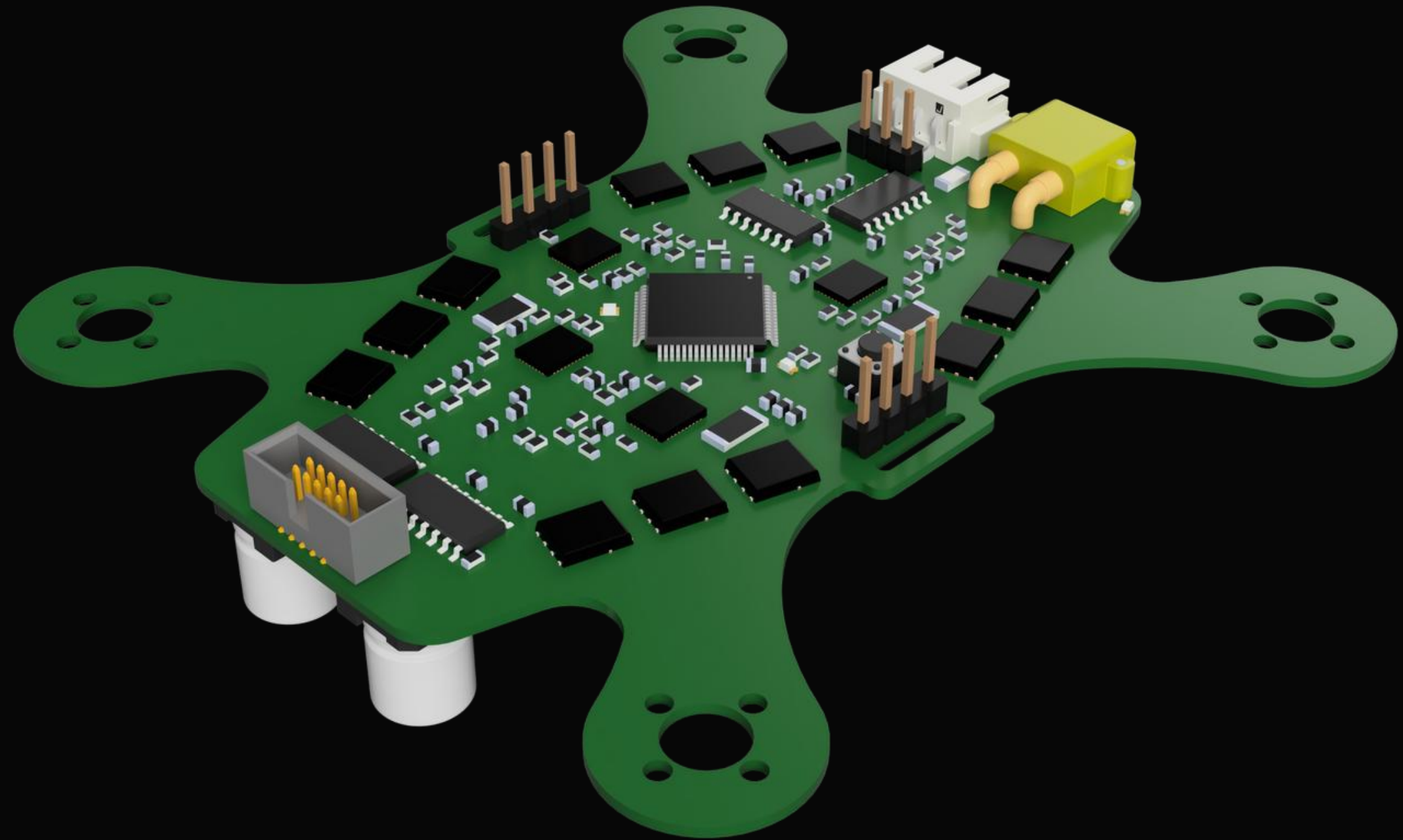


ELECTRONIC SPEED CONTROLLER

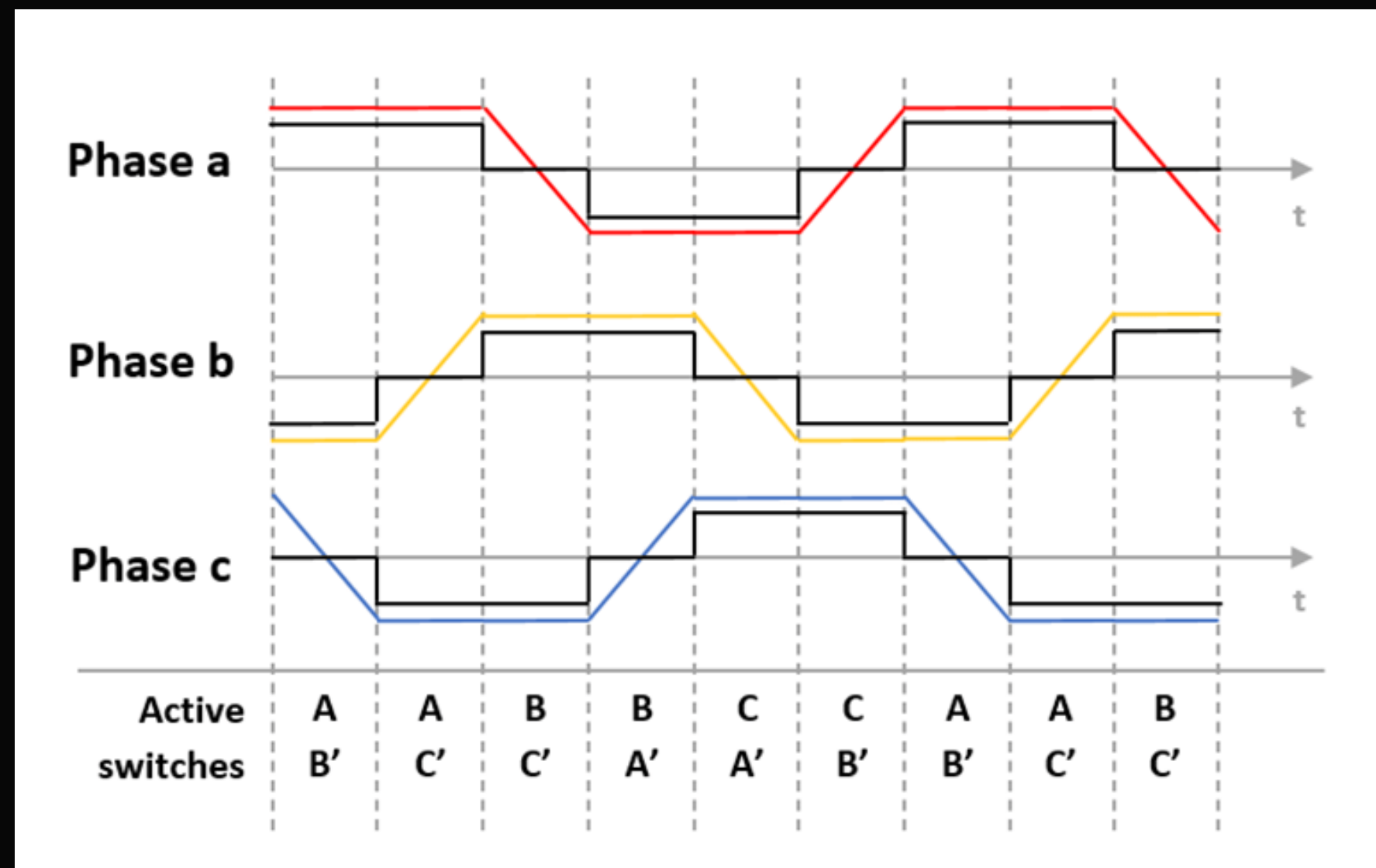
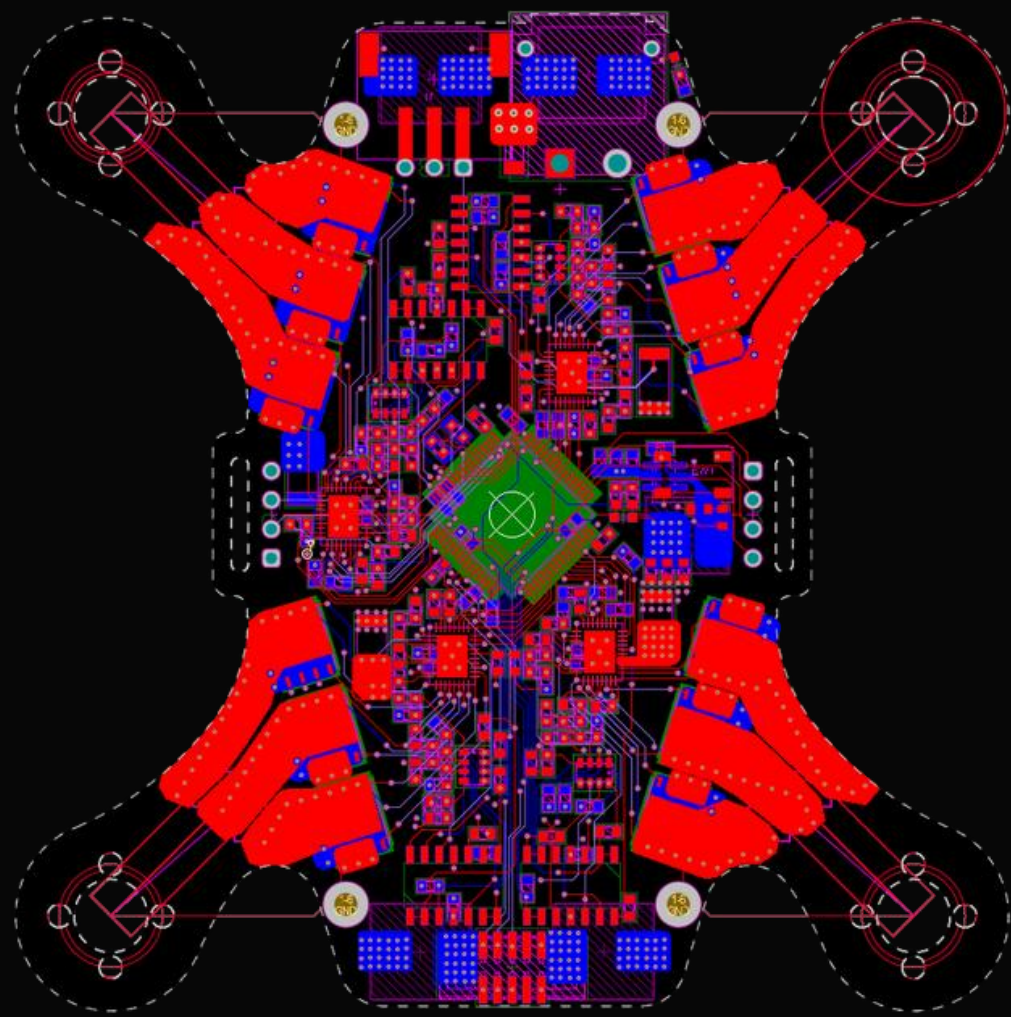


ELECTRONIC SPEED CONTROLLER

- Stall detection
- Back EMF Commutation
- Over current protection
- Motor song indicator
- Voltage Regulation



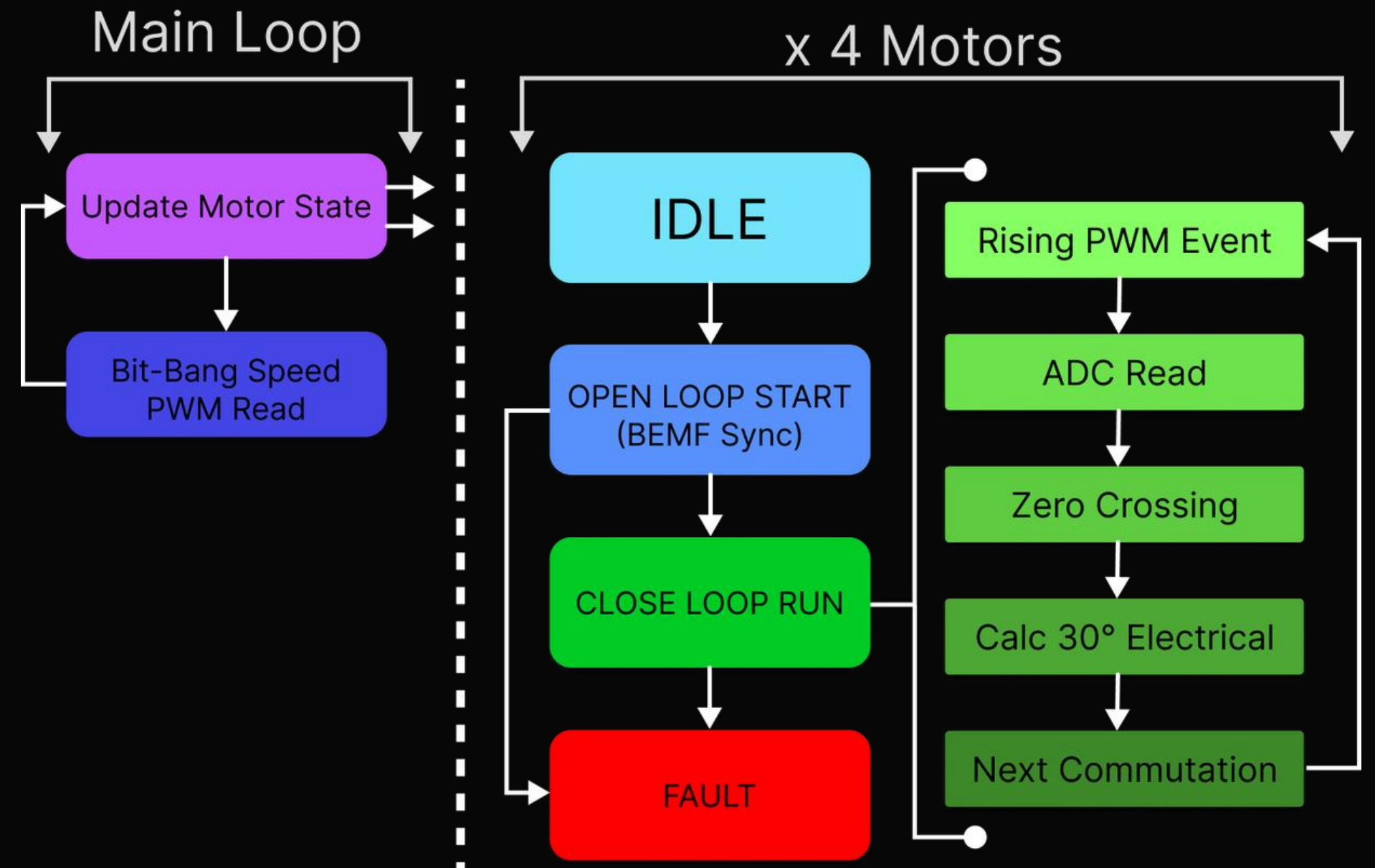
TRAPEZOIDAL SIX-STEP COMUTATION



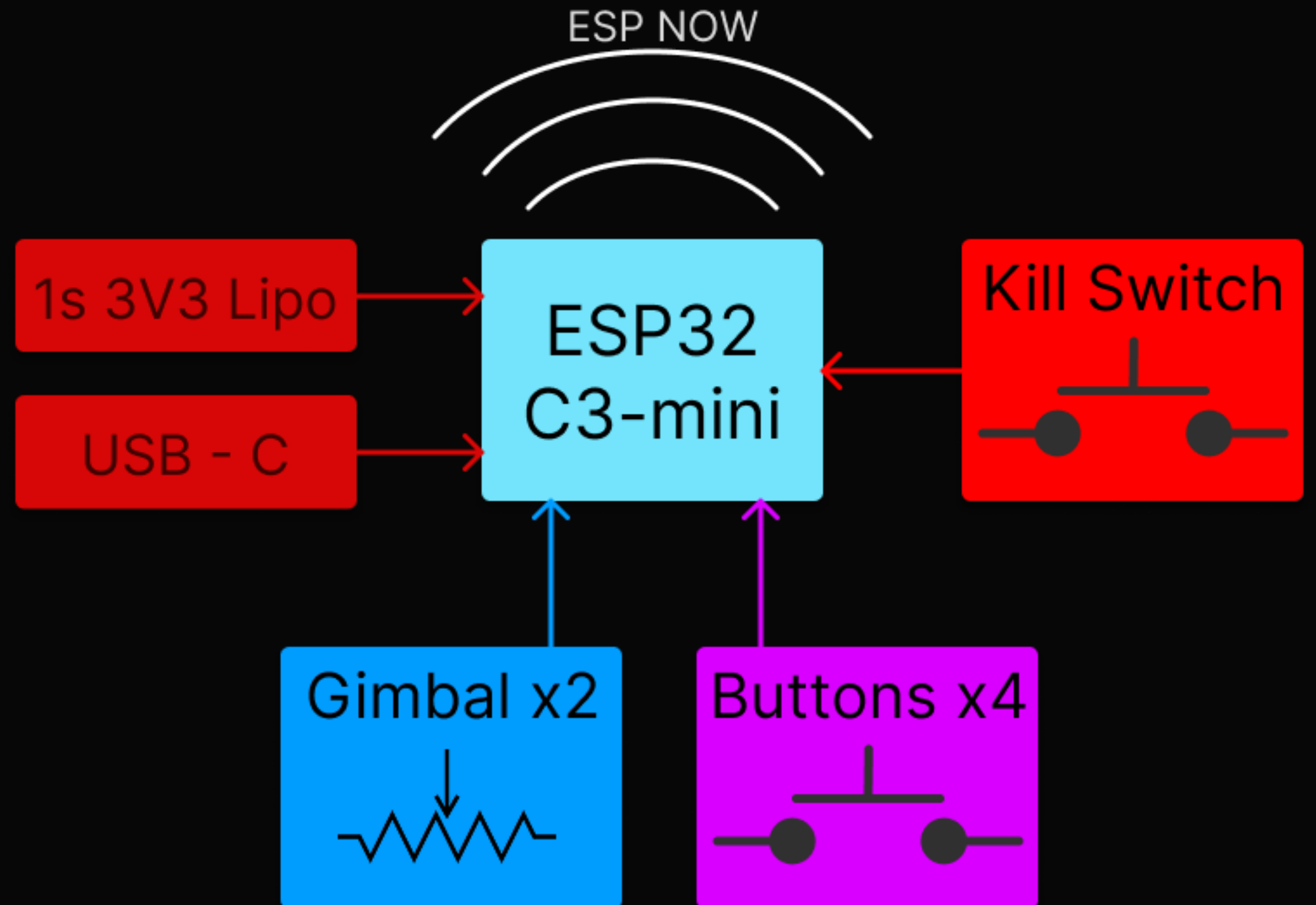
Source: MathWorks MATLAB

ESC FIRMWARE

- Each motor is generally controlled using as state machine
- Main loop updates the motor state, and reads the motor speed PWM sent from the flight controller board
- CLOSE LOOP RUN is the spinning state for the motor, relying on back EMF reading for commutation timing

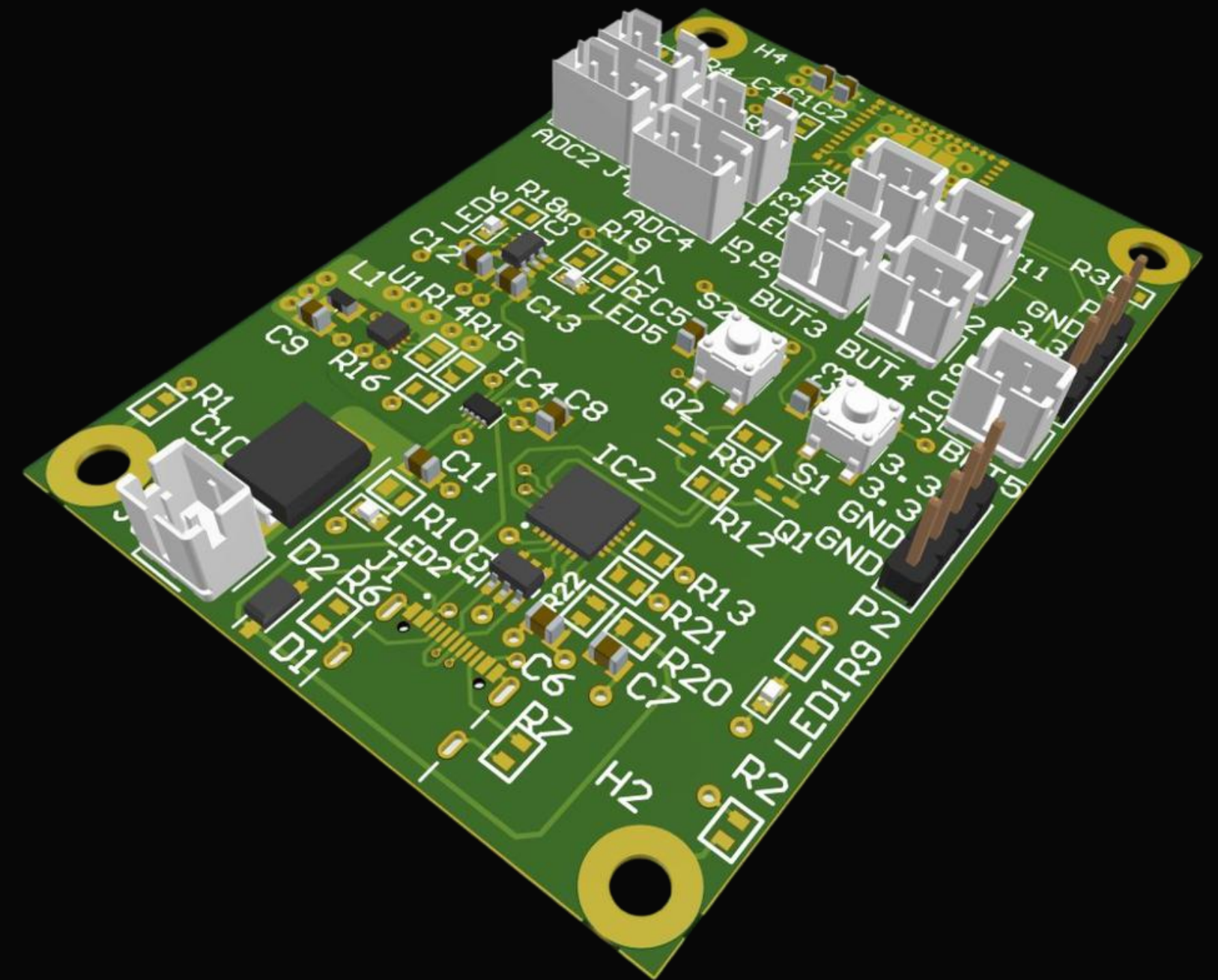


REMOTE CONTROL



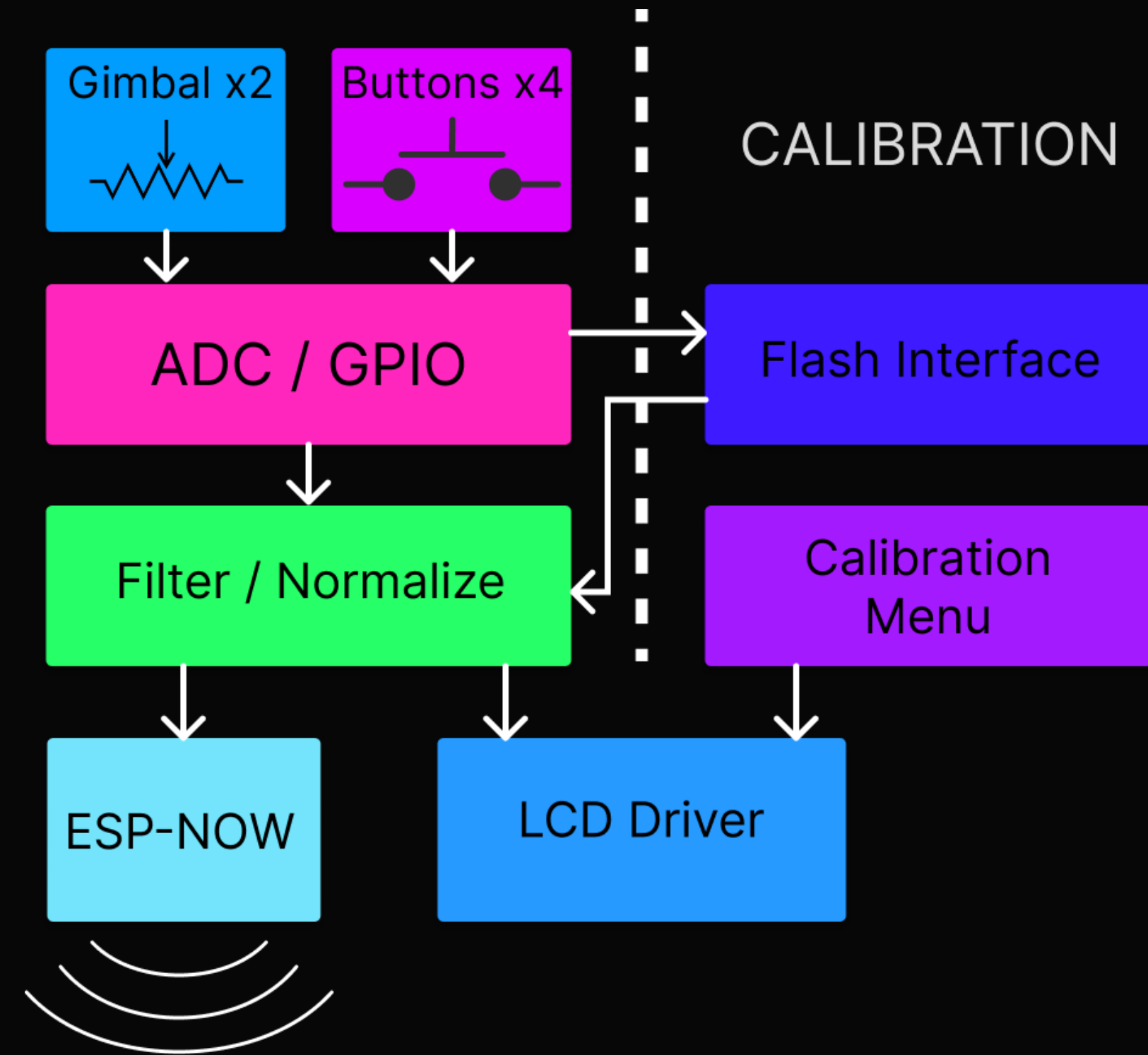
REMOTE CONTROL

- Battery or USB-C Power
- Local battery charging and power regulation
- ESP32C3-mini for ESP-NOW wireless communication
- I2C LCD user interface

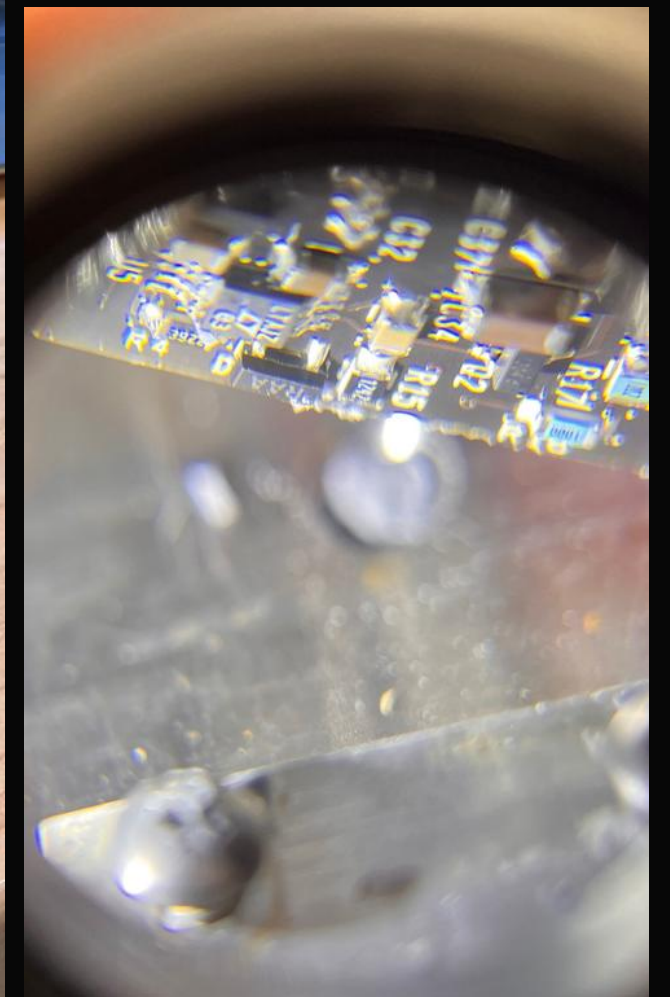
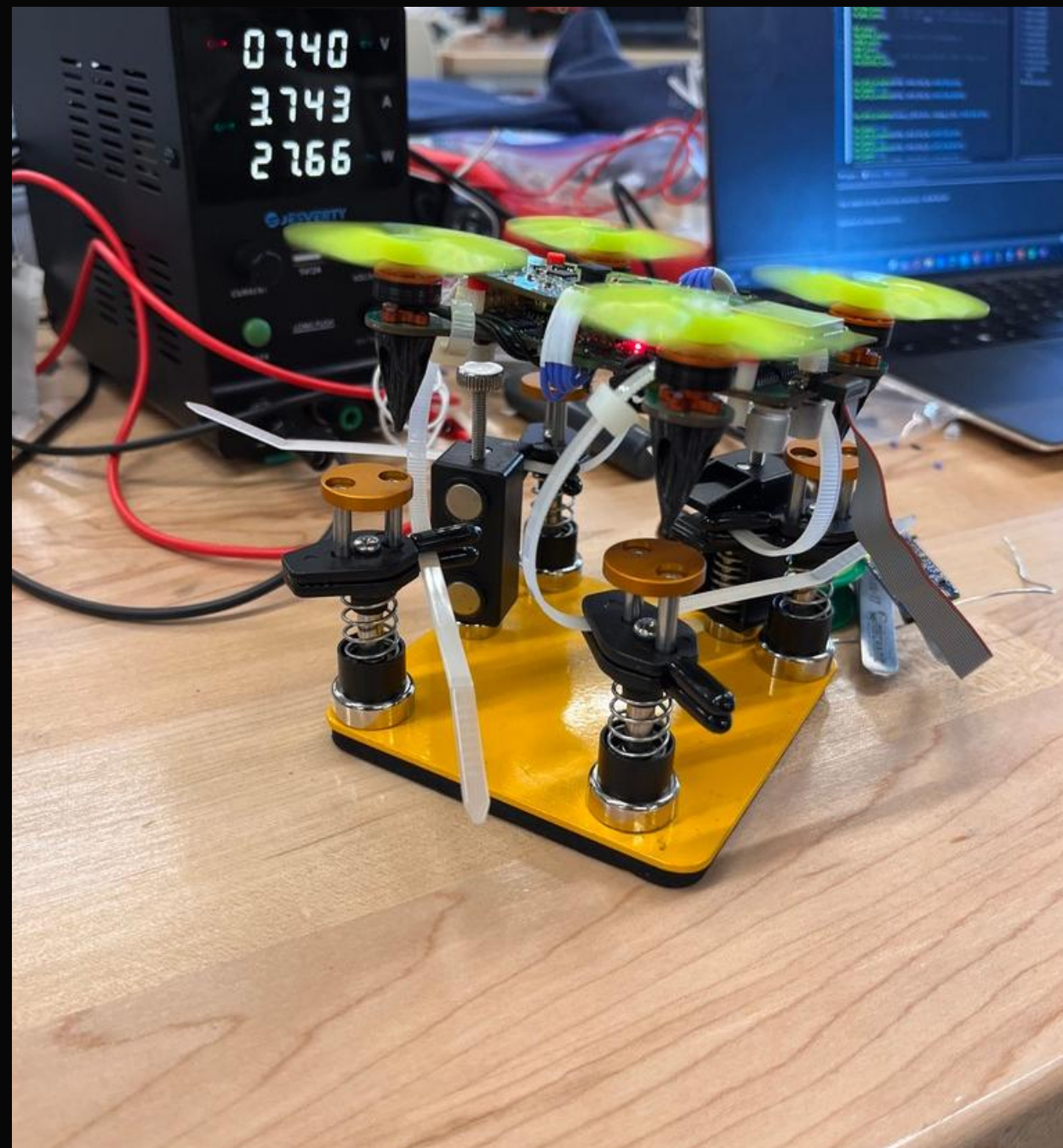
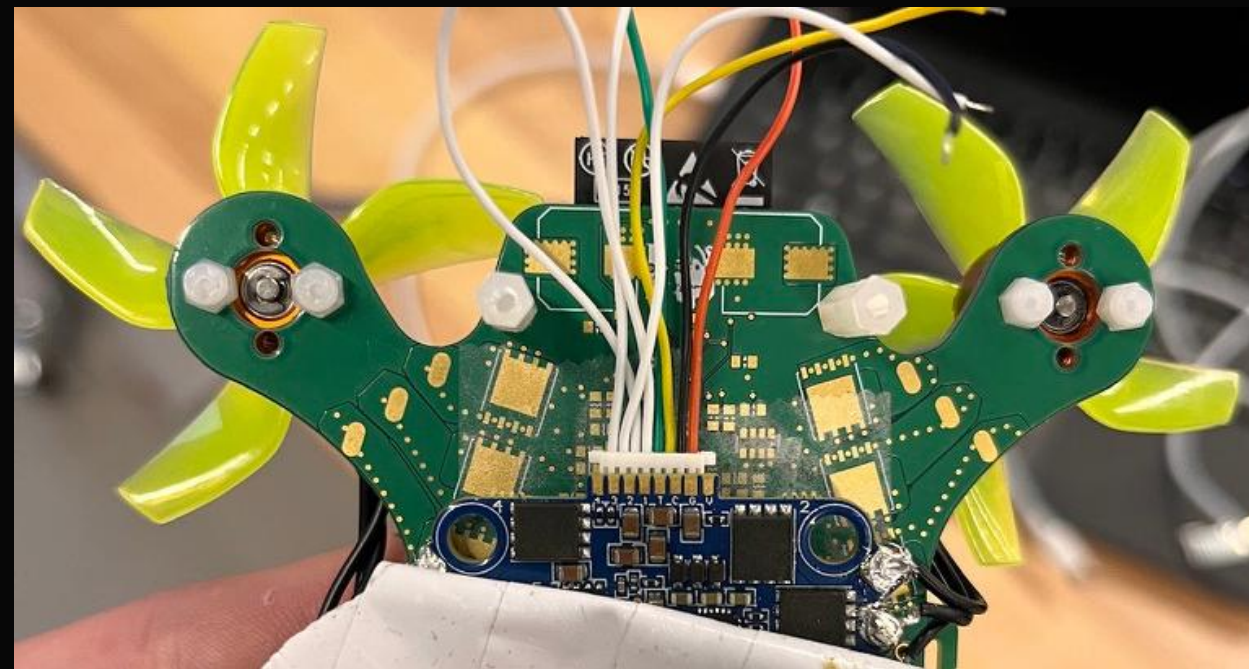


REMOTE CONTROL FIRMWARE

- Stored user interface calibration values
- IIR gimbal ADC filtering
- ESP-NOW packet queueing and acknowledgement



TESTING AND VALIDATION



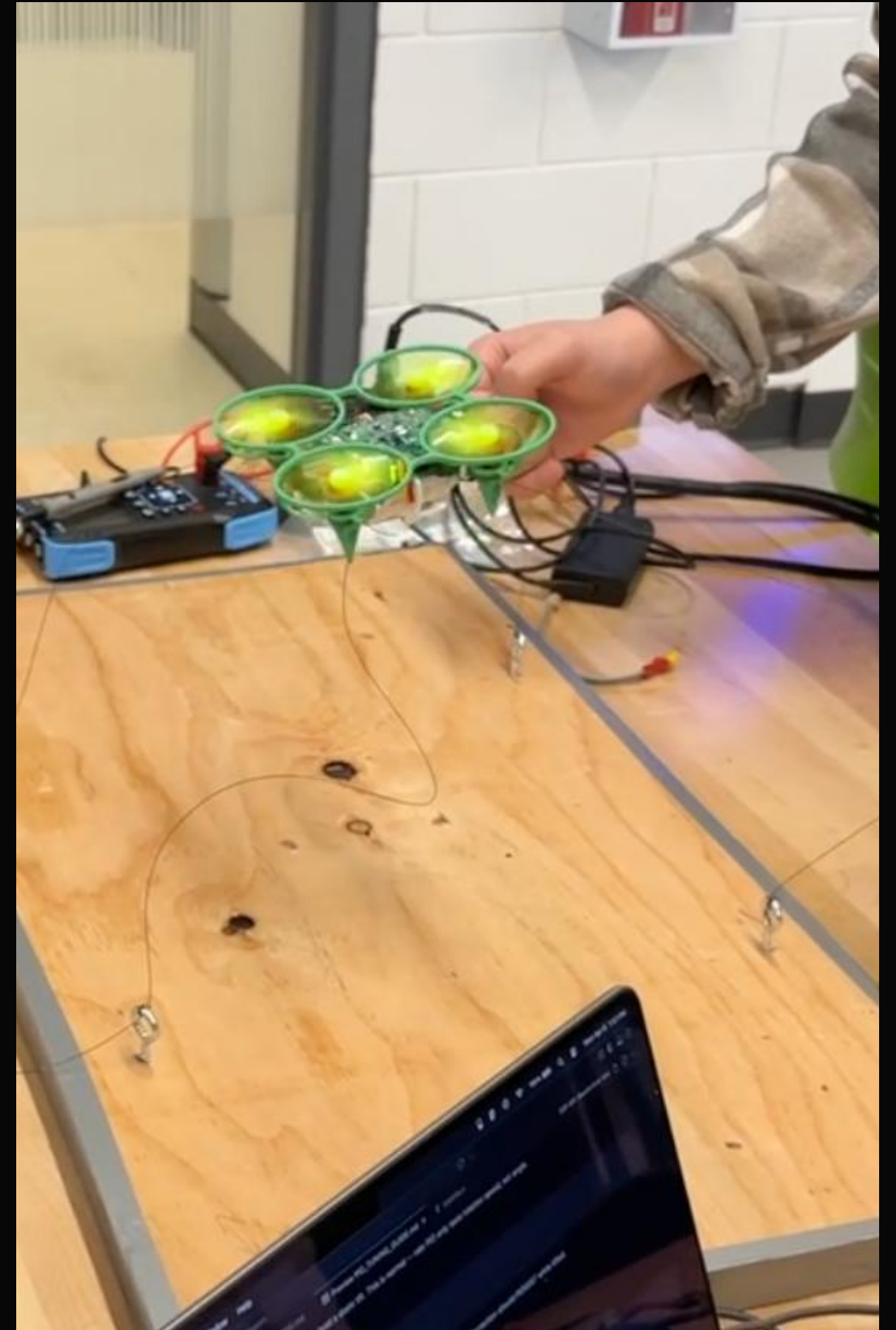
TEST SUMMARY

- 20/24 tests passed (6 UWB tests descoped)
- 1 fail: max throttle current ~18 A exceeds 15 A constraint - hover current (10.4–12 A) is within spec; transient only. BEMF efficiency is near 100%
- 2 not attempted: bare PCB hover (safety - guards kept on) and add-on module hover (not formally tested)
- Custom ESC designed, built, integrated, and flying - outperforms commercial
- Drone hovers, communicates, and stabilizes as designed
- The prototype passes acceptance testing



TESTING AND VALIDATION

- Weight: 145 g - 42% under the 250 g limit
- Thrust-to-weight: 2.32:1 - exceeds 2:1 target
- ESP-NOW range: 3% packet loss at 10 m, no consecutive drops
- Hover endurance: 5.5 minutes - nearly 3x the 2-minute requirement
- Custom ESC: all bench + flight tests passed, outperforms COTS
- PID control loop: confirmed 1 kHz, stable manual hover achieved
- Budget: \$2,265.23 - under \$2,500 cap



PROJECT MANAGEMENT AND EXPENDITURES



HOURS AND EXPENDITURES

- Total cost for R&D plus final product \$2265.23
 - Hours - ??? (a lot)
-

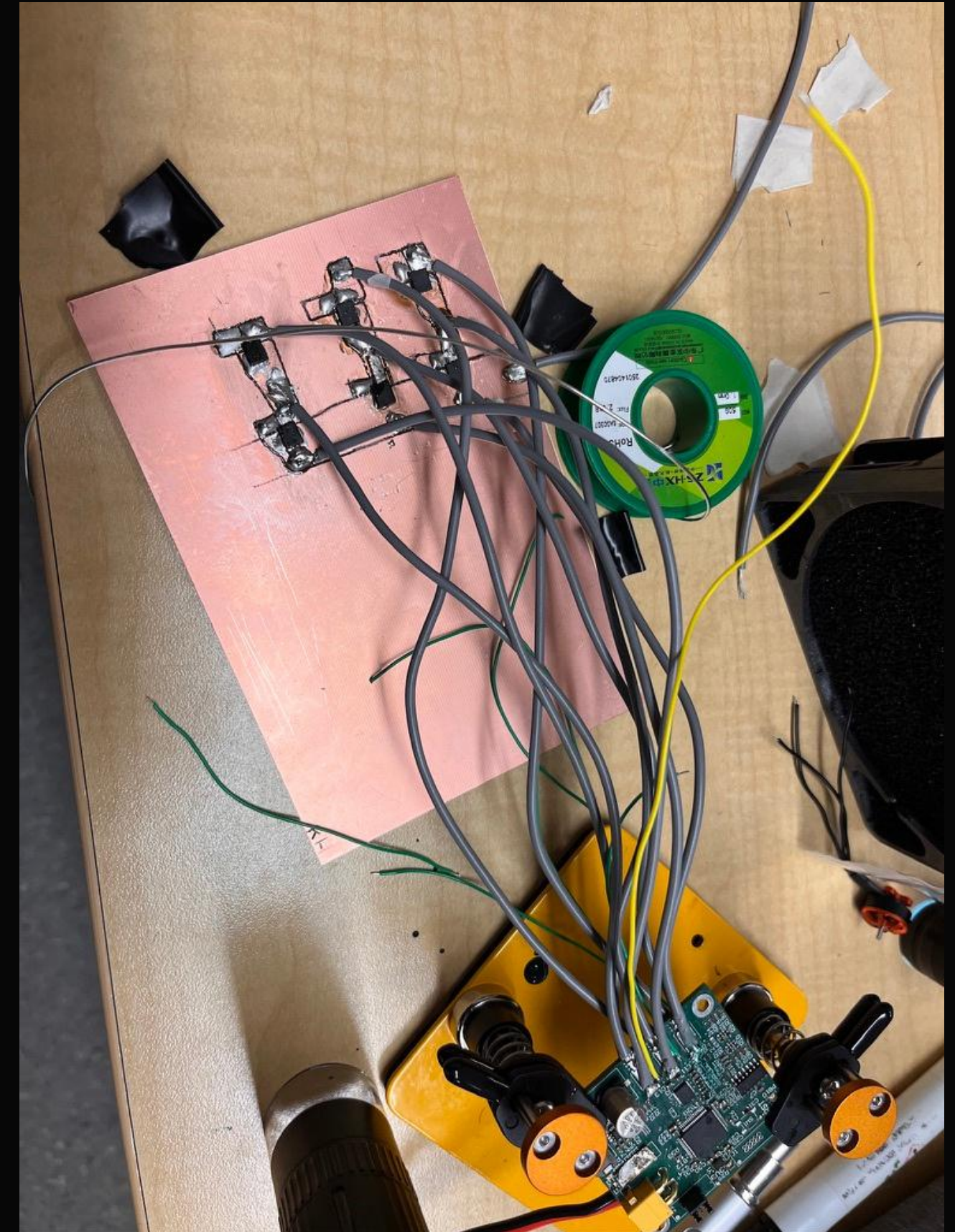


DEVIATIONS AND CHANGES

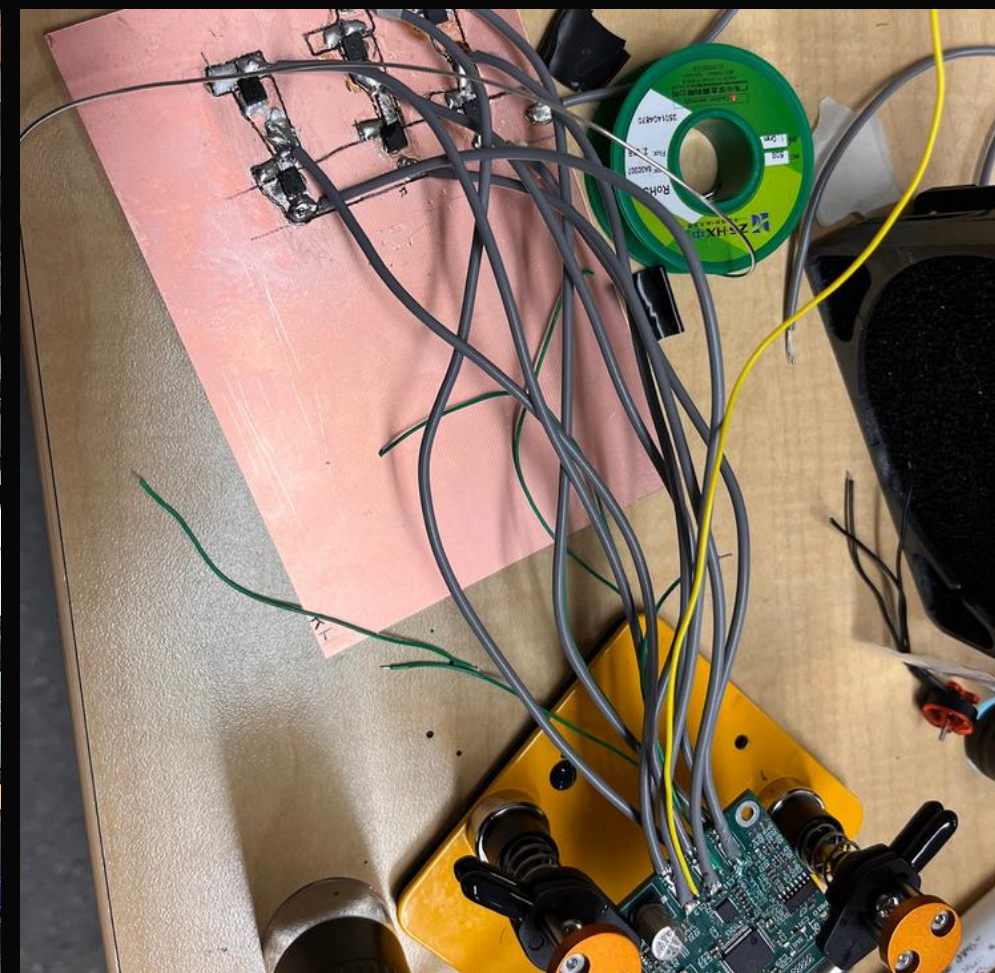
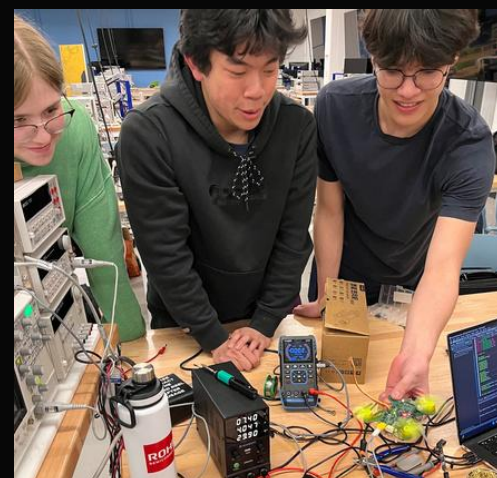
- ESC test board revision - MOSFET orientation error on v1
- UWB descope - planned as removable from day one
- Integration bottleneck - comms code ready at MVP, couldn't test until drone hardware was built

What went faster than planned

- Firmware integration - clean architecture made modules slot together easily
- PID implementation - strong upfront research accelerated development



LESSONS LEARNED



LESSONS LEARNED

Engineering Lessons

- Plan for at least one board respin - your first PCB will have errors
- Test on final hardware - dev board tuning doesn't transfer
- Invest in software architecture early - clean interfaces made integration fast
- Build redundancy + ORDER EXTRA - our backups had backups

Team-Work Lessons

- Good Git management is crucial - 6 people, multiple branches and discipline kept us sane
 - BOM management across 7 boards is chaos - overlapping components, double-orders, lost parts.
 - Scope stretch goals as removable from day one - UWB dropped cleanly when we needed to
 - Documentation always takes longer than you think - plan for it, don't leave it to the last week
-

LESSONS LEARNED

“When I breathe in the plastic, my head hurts for some reason”

Minh Do

WE NEED
TO PUT A
GROUP PIC.
I LOVE MY
FREINDS
AND I HAD
SM FUN
THIS
PROJECT.

